CURRICULUM VITAE

Obed Tawiah Nyavor

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EDUCATION

Mechanical Engineering (Ph.D. Fall, 2025 – Present
Louisiana State University Baton Rouge, LA USA

- ***Major:*** Materials Science and Engineering
- ***Academic Standing***: 4.0 (GPA)
- ***Thesis***: Development of Artificial Muscles and Multi Stimuli Actuators via Electrospinning and Computational Modelling.

Master of Engineering (ME) Spring, 2024 – July,2025
Southern University and A&M College/Louisiana State University Baton Rouge, LA USA

- ***Major:*** Materials Science and Mechanical Engineering
- ***Academic Standing:*** 4.0 (GPA)
- ***Thesis:*** Development and Investigation of Starch-based Composite for Non-structural Applications

Bachelor of Science in Engineering (BSc.) August 2018 -November 2022
Kwame Nkrumah University of Science & Technology (KNUST) Kumasi, Ghana

- ***Major:*** Materials Engineering
- ***Academic Standing:*** 75.95 (GPA)
- ***Capstone*****:** Investigation of various Anthill clay Doped with various Metal Oxides for Capacitive Energy Storage

PROFESSIONAL EXPERIENCE

Summary: Conducting federally funded research (NASA, NSF, LAMDA) on smart polymer composites with 3D/4D printability, shape memory, and self-healing capabilities. My work on artificial muscles and actuators supports critical U.S. applications in biomedical devices, robotics, and sensors. This research advances national goals in sustainable materials, space infrastructure, and healthcare innovation

- Self-healing and Shape Memory Polymers and Composites
- Electrospinning of Polymer nanocomposites.
- Computational modelling of Polymer systems using Molecular Dynamics (MD) and Density Functional Theory (DFT).

- Application of Numerical and Experimental methods in designing artificial muscles and actuators.
- Materials Characterizations using advanced techniques like MTS, XRD, SEM, FTIR, UV-VIS, ELLIPSOMETER, DMA, DSC, TGA, I/V Characterization, SEM, TEM, STEM, EBSD, FIB, EDS, Potentiodynamics, etc.

Graduate Research Assistant August 2025 – Present
Louisiana State University Baton Roge, LA USA

- Developing DFT Model to investigate UV initiated crosslinking polymerization
- Developing MD to explore extent of crosslinking on polymer network interaction and its properties (mechanical and thermal)
- Designed an electrospinning chamber for spinning nanofiber composites as artificial muscles and actuators
- Investigating the effect of carbonation on the energy storage capacity of epoxies and vitimers

Graduate Research Assistant January 2024 - July 2025
Southern University and A&M College/Louisiana State University Baton Rouge, LA USA

- Designed extrusion-based 3D printer for designing shape memory polymer and lunar regolith composites for Lunar Habitation
- Developed and characterized semi-conductive Starch-based composites for non-structural applications including insulation
- Characterized the effect of ultra sonic frequencies on the mechanical properties of vitrimers with shape recoverability

Research Assistant November 2022-November 2023
Kwame Nkrumah University of Science & Technology Kumasi Ghana

- Performed failure analysis of steels, determining the mechanism of failure and providing preventive measures.
- Investigated charge storage in reduced graphene oxide Mechanisms in Electric Double Layer Supercapacitors
- Conducted metallurgical lab sessions for **305** juniors in the Materials and Metallurgical Program

Teaching Assistant November 2022-November 2023
Kwame Nkrumah University of Science & Technology Kumasi Ghana

- ***MSE 253 - Mechanical Behavior of Materials***: Conducted tutorials and practical sessions for **250** sophomores to understand principles of Material deformation.
- ***MSE 457/METE 453 - Metal Forming Process***: Tutored **230** undergraduate seniors to demonstrate knowledge in solid-state processing of metals, such as drawing and powder metallurgy.

- ***METE 451 - Hydrometallurgical Plant Design***: Assisted **112** Metallurgical Engineering undergraduate seniors in using OnShape CAD software to design 3D manufacturing plants.
- ***MSE 362 - Materials Joining Processes:*** Conducted tutorials and practical sessions for 300 juniors to acquire knowledge and skills in joining processes like welding.
- ***MSE 354 - Physical Metallurgy of Ferrous Metals:*** Conducted tutorials for **300** juniors to acquire industry-relevant knowledge in ferrous metal production.
- ***ME 280 - Properties of Engineering Material:*** Assisted students in demonstrating knowledge in conducting mechanical tests like Tensile testing, Charpy Impact testing, and others.

Bamfo Kwakye Exams Coordinator November 2022-
College of Engineering Exams Office, KNUST November 2023
Kumasi Ghana

- Successfully managed five examinations with zero tolerance of malpractice for more than 2000 students daily.
- Ensured prompt supply of stationery and the availability of facilities.
- Developed a new online system for script collection from the examination center.

Finance and Marketing Manager February 2021 –
KNUST Notice Board February 2022
Kumasi, Ghana

- Published over 40 articles about KNUST program admission requirements and their career opportunities.
- Optimized search index parameters to improve Google's Alexa ranking from 6 million to 400,000 in 3 months.

Head of Community Research January 2022 - October
Materials Innovation Hub 2023
Kumasi, Ghana

- Innovating the field of Materials Engineering in Ghana and using materials knowledge to solve engineering-related problems
- Visited and taught ICT and other fun sciences to pupils in the basic schools.

Intern October 2021 –
Materials Engineering Laboratory, KNUST December 2021
Kumasi, Ghana

- Repaired faulty equipment like muffle furnaces and designed new manuals for their operations.
- Trained students and other researchers on safety and research methods.
- Developed a five-step treatment process of removing toxic waste in spent pot liner (SPL) and converting it to about **96 %** pure graphite

Editorial Head July 2020 – July 2021
Assemblies of God Campus Ministry (AGCM), KNUST Kumasi, Ghana

- Introduced the New Free Worship presentation software to team boosting church attendance by **12.5%** due to live to overflow
- Equipped my team with technical skills in using the MS office tools to improve their productivity and student work
- Increase the number of the Editorial Team by about **100 %** from **5 to 11** members

TECHNICAL SKILLS

Software & Programming

- ***Communication and Office skills***: MS Word, Excel, PowerPoint, Publisher
- ***Programming Languages***: Python, JavaScript, MATLAB and Simulink, C++,
- ***Statistical Software:*** Minitab, SPSS, Excel Anova, Origin Pro
- ***Database Management:*** MS Access, MongoDB, PostgreSQL
- ***Web Development:*** Python Django, ReactJS, CommerceJS, Docker, Flatter

Computational & Simulation Tools

- ***Modeling & Simulation***: Density Functional Theory with ORCA, Molecular Dynamics with LAMMPS, Xpert Highscore, VESTA, Avogadro, Digital Micrograph for S/TEM
- ***AI & ML Tools***: OpenCV, TensorFlow, PyTorch, NumPy, Pandas
- ***CAD & Design Software***: AutoCAD, OnShape, SolidWorks

Materials Characterization & Testing

- ***Electron Beam Characterization***: EBSD, SEM-EDS, TEM-EDS, STEM
- ***Nanoscale Machining:*** Focused Ion Beam
- ***Thermal Characterization:*** TGA, DMA, DSC and Rheometer
- ***Mechanical Testing:*** MTS Testing, Impact and Charpy Testing
- **Other Techniques**: FTIR, UV-VIS, Ellipsometer, Potentiodynamics, I/V Characterization

Hardware & Circuitry

- Arduino, Raspberry Pi, Troubleshooting

Research & Leadership

- **Research Management**: EndNote, Mendeley
- **Leadership**: Gallup Certified Strategic Leader

SELECTED PEER-REVIEWD JOUNAL ARTICLES

- **Obed Nyavor**, J. Konlan, K.Y. Gyabaah, P. Mensah, G Li. “Development of A Biodegradable Tapioca Starch-Based Polymeric Composite for Non-Structural Applications” – SPE Polymers (2025)

- K.Y. Gyabaah, **Obed Nyavor**, B. Mahoney, M. Jahan, P. Mensah, G. Li. "Extrusion-based Additive Manufacturing of Regolith-Filled Shape Memory Vitrimer Composite for Lunar Construction". (2025) -Under Review

CONFERENCE AND PRESENTATIONS

IMECE 2024 – ASME Conference — Nov 2024
"*Development of a Biodegradable Starch-based Polymeric composite for Non-structural Application*" – **Presenter** — Portland, OR, USA

Emerging Researchers Conference — May 2025
"Development and Investigation of Conductive Starch based Composite" – Presenter — Atlanta, GA, USA

Kumasi Centre for Collaborative Research, KNUST, Annual Conference — June 2022
"*Biophotonics; A reliable and Faster Bio-imaging and Disease Diagnosis*" – Presenter — Kumasi, Ghana

KNUST Green Energy Challenge — July 2021
"*Piezoelectricity for Sustainable Green Energy in KNUST*" - Presenter — Kumasi, Ghana

PROFESSIONAL AFFILIATION

American Society of Mechanical Engineers (ASME) – Member ID 22099754409

National Society of Black Engineers (NSBE) - Member

ASM Internal – Member

American Society of Welding Engineers

American Composite Manufacturers Association

AWARDS, FUNDING & GRANTS

- ***NSF Grant OIA-2418415***: Supported research on lunar regolith composites and shape memory polymers for lunar habitation.
- **NSF Grant OIA-1946231**: Funded development of starch-based composites and 3D printing technologies.
- **Louisiana Materials Design Alliance (LAMDA)**: Research funding for advanced polymer composites and undergraduate mentorship.
- **KNUST Research Fund**: Awarded GHC 40,000 (200% of requested) for research on spent pot liners for supercapacitor electrodes.
- **Provost's Honor Student (4x)**: Awarded for maintaining GPA > 3.75 at KNUST.

- **Most Innovative Student**: Selected among 1,200 students for outstanding innovation in materials engineering.
- **CREST Fellowship (2024 Cohort)**: Mentored students and led outreach on global challenges through STEM.
- **Conference Presenter – IMECE 2024, Emerging Researchers 2025**: Selected to present federally funded research.

REFERENCES

Would be Available upon Request